

CYBERSECURITY INTERNSHIP

Organization name: Elevate labs

TASK - 3: Networking Basics for Cyber Security

1. Basic Networking Concepts Studied

- **IP Address (Internet Protocol Address)**

An IP address is a logical numerical address assigned to each device connected to a network. It is used to uniquely identify devices and enable communication between them. There are two main types of IP addresses:

- **IPv4:** Uses a 32-bit address format (e.g., 192.168.1.1)
- **IPv6:** Uses a 128-bit address format to support a larger number of devices
In packet analysis, IP addresses help identify the source and destination of network traffic.

- **MAC Address (Media Access Control Address)**

A MAC address is a unique physical address assigned to a network interface card (NIC) by the manufacturer. It is used for communication within a local network. Unlike IP addresses, MAC addresses do not change and are used at the data link layer.

In Wireshark, MAC addresses help trace devices communicating within the same local network.

- **DNS (Domain Name System)**

DNS is a system that translates human-readable domain names (such as google.com) into machine-readable IP addresses. Without DNS, users would need to remember IP addresses instead of domain names.

DNS traffic analysis helps in identifying which websites are being accessed and is useful in detecting suspicious or malicious domain requests.

- **TCP (Transmission Control Protocol)**

TCP is a connection-oriented protocol that provides reliable data transmission. It ensures that data packets are delivered in the correct order and without loss. TCP uses error checking, acknowledgements, and retransmissions.

The TCP three-way handshake (SYN, SYN-ACK, ACK) is used to establish a secure and reliable connection between two devices.

- **UDP (User Datagram Protocol)**

UDP is a connectionless protocol that prioritizes speed over reliability. It does not perform error checking or retransmission of lost packets. Because of this, UDP is faster than TCP and is commonly used in applications such as DNS queries, video streaming, and online gaming.

In packet analysis, UDP traffic is useful for understanding fast, lightweight communication.

Understanding these networking concepts helped in correctly analyzing captured packets, identifying protocols, tracking communication flow, and understanding potential security risks during network traffic analysis.

2. Installation of Wireshark

Wireshark was installed on the system and configured successfully. After installation, the active network interface (Wi-Fi) was selected to begin capturing live network traffic.

3. Capturing Live Network Traffic

Wireshark was used to capture real-time network packets while performing normal activities such as browsing websites. Once the capture started, packets related to different protocols were displayed continuously, showing real-time communication between the system and external servers.

4. Packet Filtering by Protocol

To analyze specific traffic, protocol-based filters were applied in Wireshark:

- HTTP filter: Used to view unencrypted web traffic
- DNS filter: Used to view domain name queries
- TCP filter: Used to analyze TCP connections

Filtering made it easier to focus on relevant packets instead of all captured traffic.

5. Observation of TCP Three-Way Handshake

The TCP three-way handshake process was studied to understand how a connection is established between a client and a server. Ideally, this process consists of three steps:

1. SYN: The client sends a request to start a connection.
2. SYN-ACK: The server responds by accepting the request.
3. ACK: The client confirms the connection.

During the practical analysis using Wireshark, I was able to clearly observe ACK packets in the captured traffic. The SYN and SYN-ACK packets were not consistently visible, which may be due to ongoing or already established connections at the time of capture or background network activity.

By observing the ACK packets, it was understood that active TCP connections were already in progress and data transfer was taking place. This observation helped me understand how TCP maintains reliable communication even after a connection is established.

6. Plain-Text Traffic vs Encrypted Traffic

During packet analysis, a clear difference was observed between plain-text and encrypted traffic:

- HTTP traffic: Data could be viewed in readable plain text
- HTTPS traffic: Data was encrypted and not readable

This observation highlighted the importance of encryption in protecting sensitive data during transmission.

7. DNS Query Capture and Analysis

DNS traffic was captured and analyzed by applying the DNS filter. The analysis showed how domain names requested by the user are translated into IP addresses. This helped in understanding how systems locate servers on the internet.

8. Saving Packet Capture Files

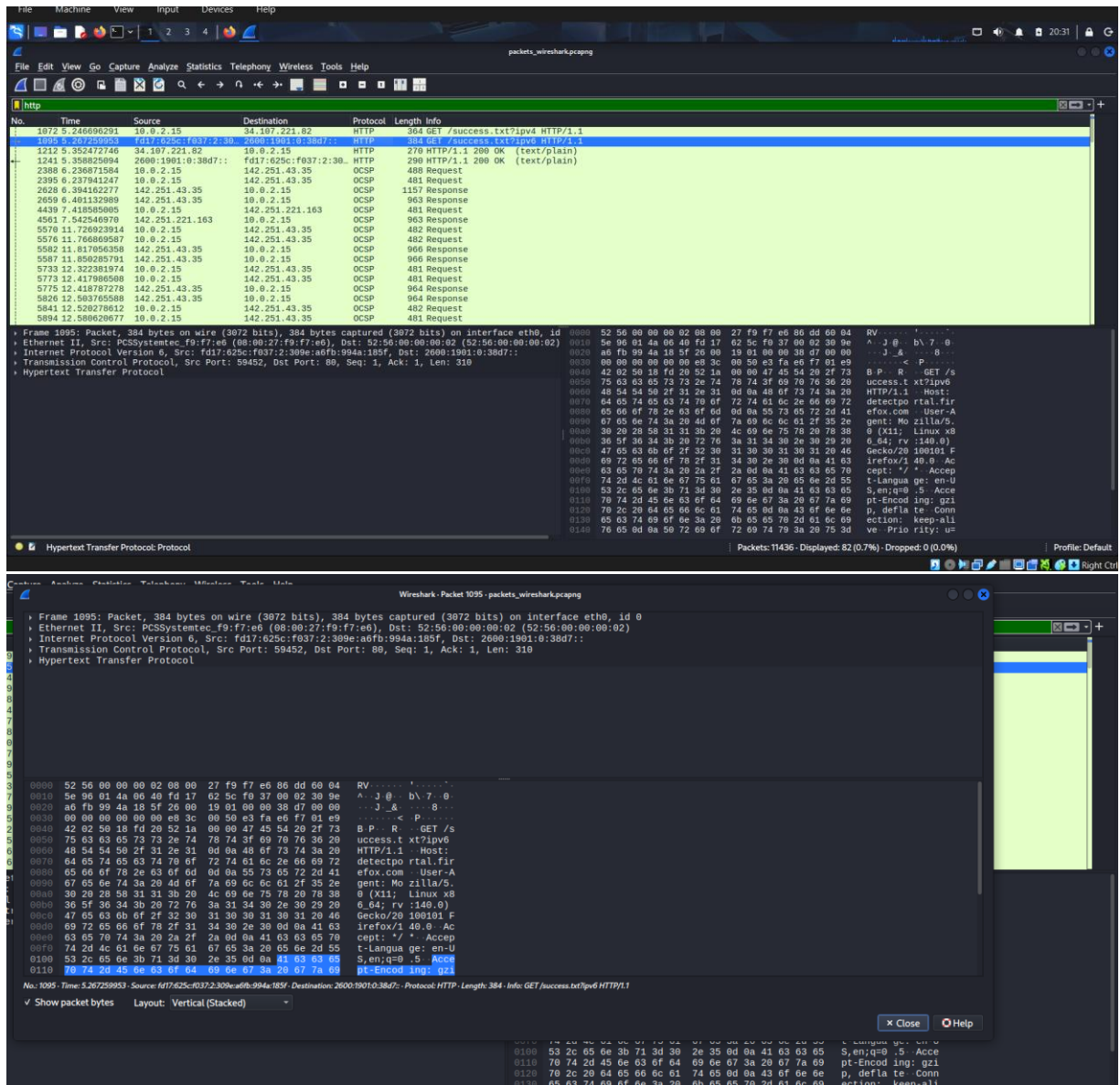
The captured network traffic was saved in .pcapng format for future analysis and documentation. This packet capture file serves as evidence of the practical work performed during the task.

9. Observations

- Different network protocols such as TCP, HTTP, HTTPS, and DNS were observed.
- TCP connections follow a structured three-way handshake process.
- Unencrypted traffic can expose sensitive data.
- DNS queries reveal the websites accessed by a user.
- Packet analysis is an important method used in cyber security to detect suspicious activities.

10. Conclusion

This task gave me hands-on experience with basic networking concepts and packet analysis using Wireshark. By capturing and studying live network traffic, I was able to understand how data moves across a network. This helped me see how network-related attacks can be identified and why monitoring traffic is important for cyber security.



packets_wireshark.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
7668	15.767838129	8.8.8.8	10.0.2.15	DNS	188	Standard query response 0xee76 AAAA fonts.googleapis.com AAAA 2404:6800:4007:80f1::200a
7673	15.793682233	8.8.8.8	10.0.2.15	DNS	137	Standard query response 0xc822 HTTPS fonts.googleapis.com SOA ns1.google.com
8765	16.669405920	10.0.2.15	8.8.8.8	DNS	77	Standard query 0x739b HTTPS fonts.gstatic.com
8770	16.693974350	10.0.2.15	8.8.8.8	DNS	77	Standard query 0x9310 A fonts.gstatic.com
8771	16.686244620	10.0.2.15	8.8.8.8	DNS	77	Standard query 0x9813 AAAA fonts.gstatic.com
8785	16.698628955	8.8.8.8	10.0.2.15	DNS	182	Standard query response 0x739b HTTPS fonts.gstatic.com HTTPS
8796	16.722543971	8.8.8.8	10.0.2.15	DNS	93	Standard query response 0xc3c1 A fonts.gstatic.com A 142.250.182.67
8797	16.722544542	8.8.8.8	10.0.2.15	DNS	105	Standard query response 0x9813 AAAA fonts.gstatic.com AAAA 2404:6800:4007:8311::2003
9800	17.323445531	10.0.2.15	8.8.8.8	DNS	79	Standard query 0xa405 A accounts.google.com
9810	17.323181374	10.0.2.15	8.8.8.8	DNS	79	Standard query 0x3f0b A accounts.google.com
9811	17.329469198	10.0.2.15	8.8.8.8	DNS	79	Standard query 0xa405 AAAA accounts.google.com
9854	17.373755841	8.8.8.8	10.0.2.15	DNS	129	Standard query response 0x5417 HTTPS accounts.google.com SOA ns1.google.com
9855	17.373756471	8.8.8.8	10.0.2.15	DNS	187	Standard query response 0xa405 AAAA accounts.google.com AAAA 2404:6800:4003:c11::54
9857	17.373756622	8.8.8.8	10.0.2.15	DNS	95	Standard query response 0x3f0b A accounts.google.com A 74.125.130.84
9990	18.657211558	10.0.2.15	8.8.8.8	DNS	74	Standard query 0xa502 HTTPS www.google.com
9991	18.658615733	10.0.2.15	8.8.8.8	DNS	74	Standard query 0xf090 A www.google.com
9992	18.658252536	10.0.2.15	8.8.8.8	DNS	74	Standard query 0x0cad AAAA www.google.com
10024	18.358097674	8.8.8.8	10.0.2.15	DNS	99	Standard query response 0xa502 HTTPS www.google.com HTTPS
10025	18.358096265	8.8.8.8	10.0.2.15	DNS	90	Standard query response 0xf090 A www.google.com A 142.250.206.68
10026	18.358098365	8.8.8.8	10.0.2.15	DNS	182	Standard query response 0x0cad AAAA www.google.com AAAA 2404:6800:4007:8301::2004

Frame 9800: Packet, 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface eth0, id 0 0000 52 55 0a 00 02 02 08 00 27 f9 f7 e6 08 00 45 00 RU.....E

Ethernet II, Src: PCSysntec_f9:f7:e6 (08:00:27:f9:f7:e6), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02) 0010 00 41 43 62 46 00 40 11 d0 20 0a 00 02 0f 00 00 Ach@.....

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 8.8.8.8 0020 00 00 0f 00 35 00 20 1c 5d 54 17 01 00 00 01S...JT....

User Datagram Protocol, Src Port: 45231, Dst Port: 53 0030 00 00 00 00 00 08 61 63 63 6f 75 6e 74 73 06A counts

Domain Name System (query) 0040 67 6f 6f 67 6c 65 03 63 6f 6d 00 00 41 00 01google c om - A

Transaction ID: 0x5417

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

[Response ID: 9854]